PRODUCT REQUIREMENT DOCUMENT (PRD)

UMHackathon 2026

umhackathon@um.edu.my

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Email: umhackathon@um.edu.my

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Project Name: AgroTech : Strategic Farmer Advisor

Version: 1.0

1. Project Overview

- **Problem Statement:** Smallholder farmers in Malaysia often face a "yield gap" and financial instability due to fragmented data and limited analytical tools. Currently, crucial decisions regarding harvest timing, replanting, and financial planning are based on manual observation or traditional intuition, which is often inaccurate. This lack of "Decision Intelligence" makes farmers vulnerable to volatile CPO prices and climate risks, hindering their economic empowerment.
- **Target Domain:** Domain 2: AI for Economic Empowerment & Decision Intelligence. Specifically, this project focuses on enabling precision agriculture and data-driven financial decision-making for the over 500,000 palm oil smallholders in Malaysia.
- **Proposed Solution Summary: AgroTech** is an AI-powered decision support system that utilizes Gemini Flash 3.1 engine to analyze fragmented data—including tree age, local weather patterns, and global supply/demand trends. The system converts complex technical and market indicators into simple, actionable English or Malay instructions. By shifting the farmer's role from a manual laborer to a data-informed manager, the system empowers them to optimize harvest cycles, plan for replanting, and protect their profit margins against market volatility.

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2. Background & Business Objective

- **Background of the Problem:** For decades, Malaysian palm oil smallholders have relied on traditional, manual methods for farm management. Harvesting decisions are often based on visual intuition of fruit color, and financial planning is done through fragmented paper records or memory. This linear, manual pathway—where a farmer observes, guesses maturity, and then reacts to market prices—lacks data-driven precision. The specialized skills required to analyze complex global supply-demand trends and long-term weather lags (12-36 months) are currently out of reach for the average farmer, blocking them from maximizing their land's economic potential.

- **Importance of Solving This Issue:** Solving this enables farmers to shift from "reactive" to "proactive" management, potentially closing the 15-25% yield gap caused by improper harvest timing. By breaking the analytical skill barrier, a farmer can visualize the long-term financial impact of their current decisions in seconds rather than years of trial and error. This speed allows for rapid pivoting during price volatility, ensuring that smallholders remain solvent and competitive against larger, tech-enabled plantations.

- **Strategic Fit / Impact: AgroTech** aligns directly with the UMHackathon 2026 goal of "Economic Empowerment & Decision Intelligence" by turning fragmented data into functional insights. It utilizes Gemini engine as a "Strategic Decision Layer," which can be integrated into existing local agriculture ecosystems (like MPOB initiatives) to facilitate a more technologically resilient Malaysian palm oil landscape.

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3. Product Purpose

3.1. Main Goal of the System: Main Goal of the System: To provide an AI-powered strategic partner for Malaysian palm oil smallholders that transforms fragmented environmental and market data into actionable decisions to close the 15–25% yield gap.

3.2. Intended Users (Target Audience):

- Independent Smallholders: Farmers managing less than 40 hectares of land.
- Koperasi (Cooperatives): Managers overseeing multiple smallholder plots.
- MPOB Extension Officers: Field officers providing advisory services to local communities
- Founder of palm oil based product

4. System Functionalities

4.1. Description: The system operates as a Strategic Decision Layer that ingests multi-modal data (images of fruitlets, weather CSVs, and global market text) to provide "Context-aware reasoning".

4.2. Key Functionalities:

- **Analysis for Advice:** Based on the user's prompt, Gemini engine will provide the best advice possible to optimize the farmer's financial prospect and decision.

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4.3. AI Model & Prompt Design

4.3.1. Model Selection: Gemini 3.1 Flash is utilized for its high-velocity "Context-aware reasoning" and its native support for multilingual processing (English and Malay). Its specific strength lies in "Explanation of decisions," translating complex environmental data into relatable strategic advice for Malaysian smallholders.

4.3.2. Prompting Strategy: We utilize a Multi-step Agentic Prompting approach. This strategy is essential because our use case involves a "chain of thought" to ensure advice is grounded in reality rather than hallucination:

- **Step 1:** Analyze unstructured user inputs (e.g., specific crop type and location-based environmental factors).
- **Step 2:** Cross-reference user data with the specified agricultural context (e.g., specific soil requirements like Clay vs. Sandy).
- **Step 3:** Perform a "trade-off analysis" between immediate harvesting needs and long-term soil/crop health based on current situational trends.
- **Step 4:** Translate the technical result into a simple "Actionable Instruction" in Malay or English that a farmer can execute immediately.

4.3.3. Context & Input Handling: The system handles "unstructured or oversized inputs" (such as detailed farmer notes or accidental large text pastes) through **Input Validation and Truncation** to maintain model efficiency.

- **Maximum Input Size:** 1,500 tokens per analysis session (optimized for fast response times in rural areas with limited bandwidth).
- **Exceeding Limits:** If inputs exceed this, the system **automatically truncates** the text to the maximum character limit allowed by the Streamlit `text_input` component to prevent API "Payload Too Large" errors.

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4.3.4. Fallback & Failure Behavior: If the system returns off-topic or "hallucinated" responses, AgroTech employs a Rule-Based Validation Layer:

- Mechanism: The system checks AI recommendations against "Basic validation" parameters (e.g., the AI cannot suggest harvesting more than the physical maximum yield of the plot size).
- Error State: If data is inconsistent, the system triggers a "Graceful Error State," asking the user to re-verify input instead of providing potentially harmful financial advice.

5. User Stories & Use Cases

- **User Stories:**

- *"As a smallholder, I want to upload a photo of my harvest area so that I can decide if today is the optimum time to hire harvesters, saving on labor costs."*
- *"As a farm manager, I want the system to explain why my yield is low this month based on weather from two years ago so that I can plan my family's expenses better."*

- **Use Cases Replanting Decision Intelligence:**

Input: *User selects their location (e.g., Alor Setar), crop (e.g., Paddy), and soil type (e.g., Clay).*

Processing: *Gemini 3.1 Flash analyzes these environmental inputs against established agricultural best practices and real-time trends.*

Actionable Insight: *The system performs a "Situation Report" analysis, determining the most critical risk factor for that specific crop-location-soil combination.*

Output: *"📍* ***Situation Report****: Heavy Rain detected for Paddy in Clay soil.* ***Strategic Advice****: Check drainage channels immediately to prevent waterlogging and root rot. Monitor for fungal infections."*

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6. Features Included (Scope Definition)

- Bilingual Command Center: Support for actionable instructions in English and Bahasa Melayu.
- Financial Forecasting: 12-month cash flow projections based on global CPO trends.

7. Features Not Included (Scope Control):

- **Direct E-Wallet/Payment Integration:** The system provides decision intelligence but does not process financial transactions.
- **Automated Harvesting Machinery:** The system advises the human; it does not control physical hardware or robotics.

8. Assumptions & Constraints

- LLM Cost Constraint: We estimate a cost of RM0.15 to RM0.25 per user session. To keep costs "manageable at scale," we use Local Metadata Caching for global CPO prices, ensuring the system doesn't re-query the GLM for the same market data for every individual farmer in the same hour.
- Technical Constraints: The "Decision Intelligence" is delivered via a Web-App optimized for low-bandwidth rural areas. It does not support offline real-time processing; an active internet connection is required to access the Gemini GLM cloud.
- Performance Constraints: To maintain speed, image analysis for maturity indexing is limited to 5 uploads per day per user.
- User Input: The system is a "Decision-support tool," not an autonomous manager. It requires human "Basic validation" and approval before any suggested financial actions (like purchasing fertilizer) are finalized.

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9. Risks & Questions Throughout Development

- **Decision Homogenization (vs. Design Similarity):**
  - **Risk:** How do we ensure the AI doesn't give identical advice to all farmers, potentially causing local market gluts (e.g., advising everyone to harvest on the same Tuesday, overwhelming local collection centers)?
  - **Mitigation:** Gemini 3.1 Flash uses "Context-aware reasoning" to factor in the specific **Location**, **Crop Variety**, and **Soil Type** of each unique user. This ensures that "Recommendation of actions" are hyper-localized rather than generic, spreading out harvest and maintenance cycles based on specific plot conditions.

- **Data Integrity & "Hallucination" Stability (vs. Frame Stability):**
  - **Risk:** Does the user need to manually override the system if the AI generates a forecast that contradicts local observations or actual farm conditions?
  - **Mitigation:** The system includes a "Basic validation" step where the farmer reviews the AI's situational interpretation (the "Situation Report") before acting on the strategic advice. This acts as a human-in-the-loop "Decision-making" safety net to ensure the AI hasn't hallucinated environmental factors.

- **Multi-modal Interpretation (vs. Drawing Interpretation):**
  - **Risk:** What will happen if the AI is unable to contextualize non-standard agricultural data or highly specific local terminology provided by the farmer in the text inputs?
  - **Mitigation:** If the model cannot reach a high confidence level in its "Interpretation of unstructured data," it is programmed to trigger a "Fallback" state. Instead of guessing, it will ask the farmer for clearer manual text input or specific details about their crop status to ensure the "Explanation of decisions" remains accurate and safe.

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9. References

This project, developed by **Team Error 404**, adheres to the official submission guidelines of UMHackathon 2026 regarding the disclosure of external resources and artificial intelligence tools. The following resources were utilized during the developmental period:

- **Generative AI Reasoning Engine:** * **Google Gemini 3.1 Flash API:** Utilized as the central "Reasoning Service" to perform context-aware analysis of agricultural data and generate strategic instructions in both English and Malay.
    - **Developmental Assistance:** Large Language Models were used for code optimization, content generation, and technical documentation assistance.

- **External Data Sources & APIs:**
    - **OpenWeather API:** Integrated to fetch real-time environmental data (temperature, precipitation, and weather conditions) for specific Malaysian locations.
    - **Malaysian Agricultural Market Indices:** Publicly accessible price trends for Paddy, Chili, and Tomato were referenced to provide economic insights.

- **Software Frameworks & Libraries:**
    - **Streamlit:** Used as the primary cloud-native infrastructure for the user interface and frontend deployment.
    - **Python Libraries:** Standard libraries including google-generativeai and requests were used for API integration and data siphoning.

- **Compliance Statement:**
    - All external tools and datasets used fulfill the criteria of being publicly accessible and properly cited.
    - This submission is exclusive to UMHackathon 2026 and has not been entered into any other concurrent events.
    - No sensitive, identifiable personal information was used or shared within the datasets or problem statement responses.

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